Mth641 short notes prepared by Ghulam Siddique ahmadani

Metric Spaces:

Formal definition:

A metric space is a pair (X, d), where X is a set and d is a metric on X (or distance function on X), that is, a function defined on $X \times X$ such that for all $x, y, z \in X$ we have following four properties.

1*st* Property:	M_1	d is real-valued, finite and non-negative
2*nd* Property:	M_2	$d(x, y) = 0$ if and only if $x = y$
3rd Property:	M_3	$d(x, y) = d(y, x)$ (Symmetry)
4th Property:	M_4	$d(x, y) \leq d(x, z) + d(z, y)$ (Triangle Inequality)

The above four properties called axioms of metric space. As metric space is ordered pair so we take $X \times X$ mean two elements from set X.

Subspace:

Formal definition:

A subspace (Y, d) of (X, d) is obtained if we take a subset $Y \subseteq X$ and restrict d to $Y \times Y$. Thus the metric on Y is the restriction

$$\tilde{d} = d|_{Y \times Y}$$

$\tilde{d}$ is called the metric induced on Y by d.

Three dimensional Euclidean Space $\mathbb{R}^3$

$\mathbb{R}^3 = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ $\mathbb{R}^3$ plane

In $\mathbb{R}^3$ set the elements are in ordered triple form whose all entries are real numbers. Suppose u and v be two points in $\mathbb{R}^3$ such as

$u = \{\xi_1, \xi_2, \xi_3\}$ *and* $v = \{\eta_1, \eta_2, \eta_3\}$, $\xi_i, \eta_i \in \mathbb{R}$,
(*where* ξ *is exai and* η *is eta*)

The distance between *u* and *v* is

$$d(u, v) = \sqrt{(\xi_1 - \eta_1)^2 + (\xi_2 - \eta_2)^2 + (\xi_3 - \eta_3)^2}$$

Thus $(\mathbb{R}^3, d)$ satisfy all four properties of metric space and is a metric space.

n-Tuples Euclidean Space $\mathbb{R}^n$

Unitary Space C^n

$C^n = \{\xi_1, \ldots\ldots\ldots \xi_n) \mid \xi_i \in C\}$ *wherer C is complex n*

(note: In C^2 both the first and second elements are from complex numbers as $(1+i, 1-i)$ and also in C^n all the n tuples from $\xi_i, \eta_i \in C$ to ξ_n are all complex elements).

Let $z = a + ib$, $w = c + id$ then

$|z| = \sqrt{a^2 - b^2}$, $|w| = \sqrt{c^2 - d^2}$ and

as $z-w=(a-c)+i(b-d)$, $|z-w| = \sqrt{(a-c)^2 + (b-d)^2}$

Suppose u and v be two points in C^n such as

$u = \{\xi_1, \xi_2, \ldots\ldots \xi_n\}$ *and* $v = \{\eta_1, \eta_2, \ldots\ldots\ldots, \eta_n\}$, $\xi_i, \eta_i \in \square$,

(*where* ξ *is exai and* η *is eta and are complex numbe*)

Sequence Space l^∞ :

As a set X we take the set of all bounded sequences of complex numbers. These bounded sequences may be real or complex but we take here complex numbers. Collection of all complex number.

If we take all sequences of complex number which are bounded in a set then the set is called l^∞ .

i): Function Space C[a, b]

As a set X we take the set of all real-valued functions x, y,.... .... Which are functions of an independent real variable t. And are define and continuous on a given closed interval

$$J=[a, b]$$

ii): Discrete Metric Space

In discrete metric space let X be a set, which could be real number, R^3, R^n , function or set of sequence etc. then we need a distance function.

Distance function is generalized, that if we take two elements from X and those two elements are same then its distance is zero.

Sequence Space s:

The previous example consists of only bounded set but this space consists of all (bounded or unbounded) sequences of complex numbers.

Here the distance function is changed from previous one, the metric d defined by

$$d(x, y) = \sum_{j=1}^{\infty} \frac{1}{2^j} \frac{|\xi_j - \eta_j|}{1 + |\xi_j - \eta_j|}$$

Space l^p

The real Space l^p

If the elements ξ_j *and* η_j are not complex numbers but from real numbers then the space is called real space l^p.

The complex Space l^p

If the elements ξ_j *and* η_j are complex numbers then the space is called complex space l^p

Above both have the same condition that summation of $|\xi_1|^p + |\xi_2|^p + \ldots\ldots\ldots\ldots\ldots$ that should be converges and $\sum_{j=1}^{\infty} |\xi_1|^p < \infty$, and the distance function is one by one entry difference with power p and overall power 1/p.

The Hilbert Sequence Space l^2

Now the case p=2 (fixed)

The Hilbert sequence space l^2 with the metric defined by

$$d(x, y) = \left(\sum_{j=1}^{\infty} |\xi_j - \eta_j|^2 \right)^{\frac{1}{2}}$$

$$= \sqrt{\sum_{j=1}^{\infty} |\xi_j - \eta_j|^2}$$

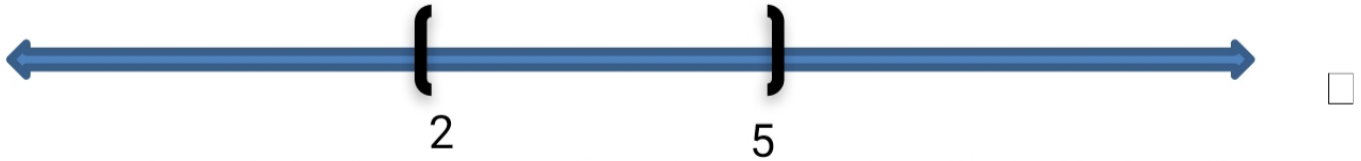
Open Set on Real Line

On real line we have open set not open ball.



(2, 5) is an open set. It includes all values between 2 and 5 but does not include 2 and 5.

Closed Set on Real Line



[2, 5] is a closed set. It includes all values between 2 and 5 including 2 and 5.

Points on the Boundary:

$d(x_o, x) = r$; *x is* on *boundary*

All the points are lies on the boundary if

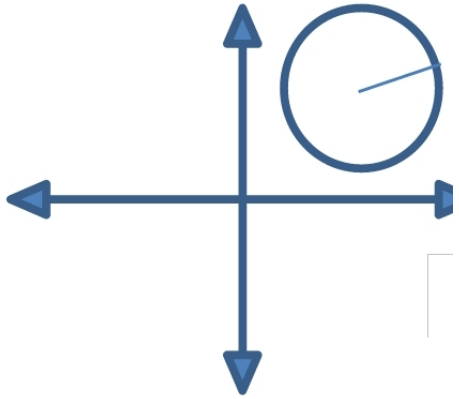
the difference of that point *x* from the center x_o is r.

$$x_o - x = r$$

x is a boundary point.

Points inside the Boundary

All the points are lies inside the boundary if



the distance between x and the center x_o

Points Outside the Bounday

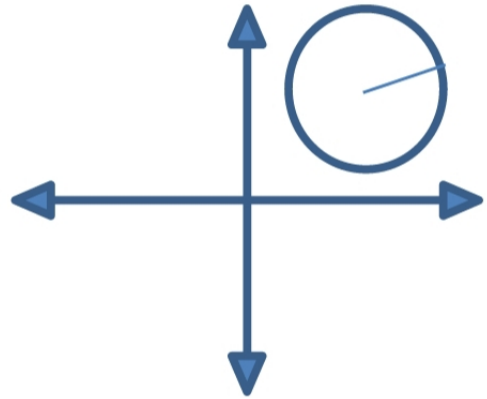
All the points are lies outside the boundary if

the distance between x and the center x_o

(i-e. the difference of that point x from the center x_o) is greater than r.

$$d(x_o, x) > r$$

x lies outside the boundary.



Open Ball:

In $\square^2$ if boundary is not included then it is open ball. It means all points inside the boundary are included. $\Rightarrow \quad d(x_o, x) < r$

Closed Ball:

In $\square^2$ if boundary is included then it is Closed ball. It means all points inside the boundary and on the boundary are included. $\Rightarrow \quad d(x_o, x) \leq r$

Ball and Sphere (General Form)

Open Ball: $B(x_o; r) = \{ x \in X \mid d(x, x_o) < r \}$

Closed Ball: $\tilde{B}(x_o; r) = \{ x \in X \mid d(x, x_o) \leq r \}$

Sphere: $S(x_o : r) = \{ x \in X \mid d(x, x_o) = r \}$

In discrete space, we have defined distance function. sphere can be empty.

$d(x, x) = 0$

$d(x, y) = 1 \quad ; \quad x \neq y$

$d(x, x_c) < r$

$d(x, x_c) \leq r$

Sphere = ϕ

For boundary we can subtract open ball from closed ball.



$$S(x_0, r) = \tilde{B}(x_0, r) - B(x_0, r)$$

Open Set:

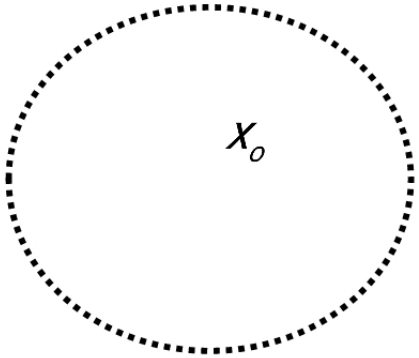
A subset M of a metric space X is said to be open if it contains a ball about each of its points.

Closed Set:

A subset K of X is said to be closed if ***its complement (in X) is open.***

that is $K^c = X - K$ is open.

Neighbor hood of a Point:



We can find an open ball around each point in Open set.

"An open ball $B(x_o, \varepsilon)$ of radius ε is often called an ε-neighborhood of x_o."

By a neighborhood of x_o, we mean any subset of $\boldsymbol{X}$ which contains an ε-neighborhood of x_o.

Radius r .

For radius r means larger values, 0.1, 0.5, 10, 40 while radius ε means very small values like 0.002, 0.000003 etc.

Radius ε .

If we take a point x then all the points around it make a ball whose radius is very small ε or ε - neighborhood of x_o.

Interior Point:

We call x_o an interior point of a set $M \subseteq X$ if M is a neighborhood of x_o.

The interior of M is the set of all interior points of M.

Int(M) is open and is the largest open set contained in M.

Collection of all open balls is an open set whether the radius ε of open ball is greater or smaller.

Topological Space:

Definition:

Let τ be collection of all open subsets of X. Then (X, τ) is said to be a topological space if it satisfies following properties:

T1): $\phi \in \tau$ *and* $X \in \tau$

T2): The union of any member of τ is a member of τ.

T3): The intersection of finitely many members of τ is a member of τ.

It holds.!

Accumulation Point (limit point):

Definition:

If M is a subset of a metric space X then x_o is a limit point of M. if it is the limit of an eventually non-constant sequence (a_i) of points of M (or limit point of M) if every neighborhood of x_o contains at least one point $y \in M$ distinct from x_o.

Translation in the form of metric space:

Let M be a subset of a metric space X, then a point x_o of X (which may or may not be a point of M) is called an accumulation point of M (or limit point of M) if every neighborhood of x_o contains at least one point $y \in M$ distinct from x_o.

All those point which are on the boundary of this open ball are accumulation points or limit points.

Closure of M:

The set consisting of the points of M and the accumulation points of M is called the closure of M and is denoted by $\bar{M}$.

Dense Set :

Definition:

A subset M of a metric space X is said to be dense in X if $\bar{M} = X$

Closure set is a set along with its limits points.

Separable Space

A metric space X is said to be separable if it has a countable subset which is dense in X.

It has two conditions; First its subset is dense and second is countable.

Separable Spaces:

A metric space X is said to be separable if it has a countable subset which is dense in X.

Examples (Separable Spaces):

- ***The Real Line*** □
- ***The Complex Plane*** □
- ***Discrete Metric Space***

The Complex Plane

In complex plane the numbers are in the form of $\{a+ib \mid a,b \in \mathbb{R}\}$, $i=\sqrt{-1}$

Or same as $\{(a,b) \mid a,b \in \mathbb{R}\}$

Discrete Metric Space:

In discrete metric space if elements are same then distance is 0 and if elements are different then distance is 1. There is no condition on set. Set can be any set.

Example 2:

Space l^p

Space mean “a set”, in this set elements are sequences, which may be real (called real space l^p) or complex numbers (called complex space l^p). Then we have define its metric, a metric means a distance function.

Now we have to show that a subset of l^p is dense and also countable then l^p is a separable space.

Bounded Sequence:

Definition:

We call a nonempty subset $M \subset X$ a bounded set if its diameter

$\delta(M) = \sup_{x,y \in M} d(x,y)$ is finite.

Here we check all distance pairs for each point against all other points, lineup all those distances and take the supremum distance point we call the diameter of that set.

It means supremum of all distance is finite then the set is bounded.

A sequence (x_n) in X is bounded sequence if the corresponding point set is a bounded subset of X.

Hence bounded sequence means finite diameter and if diameter is infinite then sequence is unbounded.

Sequence is a function whose domain is natural numbers.

Convergence of a sequence:

A sequence (x_n) in a metric space $X = (X, d)$ is said to converge or to be convergent if there is an $x \in X$ such that

$$\lim_{n\to\infty} d(x_n, x) = 0$$

x is called the limit of (x_n) and we write

$$\lim_{n\to\infty} x_n = x$$

or simply $x_n \to x$

Convergence of a sequence: (Another Definition)

We say that (x_n) converges to x or has the limit x, if (x_n) is not convergent, it is said to be divergent.

Cauchy Sequence:

Definition:

A sequence (x_n) in a metric space $X = (X, d)$ is said to be Cauchy (or fundamental) if

for every $\varepsilon > 0$ there is an $N = N(\varepsilon)$ such that $d(x_m, x_n) < \varepsilon$ for every m, n>N.

Equivalent notation $d(x_m, x_n) \to 0$ as $m, n \to$

Completeness:

Definition:

The space X is said to be complete if every Cauchy sequence in X converges that is, has a limit which is an element of X.

Theorem Convergent Sequence:

Theorem:

Every convergent sequence in a metric space is a Cauchy sequence.

Proof:

Theorem (Closure, Closed Set):

Closure is a collection of limit points and the set itself.

Limit point is such point that if we draw an open ball around it then we can find another point other than that point which belongs to that set.

A set is called a ***closed set*** if all the limits points are present in that set.

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Theorem:

Let M be a nonempty subset of a metric space $d(X, d)$ *and* $\bar{M}$ its closure as defined before then,

a): $x \in \bar{M}$ if and only if there is a sequence (x_n) in M such that $x_n \to x$.

b): M is closed if and only if the situation $x_n \in M$, $x_n \to x$ implies that $x \in M$.

Theorem (Complete Subspace):

Theorem:

A subspace M of a complete metric space X is itself complete if and only if the set M is closed in X.

As this condition is if and only if so vice versa. From previous theorem we have

Theorem (Continuous Mapping):

Theorem:

A mapping $T: X \to Y$ of a metric space (X, d) into a metric space $(Y, \tilde{d})$ is continuous at a point $x_o \in X$ if and only if $x_n \to x_o$ implies $Tx_n \to Tx_o$.

Proof:

Lemma a:

Every Cauchy sequence in a metric space is bounded.

This is for every metric space.

Lemma b:

If a Cauchy sequence has a subsequence that converges to $\bar{x}$, then the sequence converges to $\bar{x}$.

Proposition:

Every sequence of real numbers has a monotone subsequence.

Proof:

Theorem:

- is a completer metric space, i.e., every Cauchy sequence of real numbers converges.

Proof:

Let $\{x_n\}$ be a Cauchy sequence.

Remark *a* implies that $\{x_n\}$ is bounded. Now if the given Cauchy sequence is bounded then its subsequence is also bounded.

Every subsequence of $\{x_n\}$ is bounded.

Also $\{x_n\}$ has a monotone subsequence. Now $\{x_n\}$ is monotone as well as bounded.

Monotone Convergence Theorem:

If a sequence $\{x_n\}$ is monotone and bounded this implies that it is convergent.

This implies that subsequence is convergent. Now using remarks 2 if we have a Cauchy

Example:

The Euclidean space $\mathbb{R}^n$ is complete.

Proof:

Isometric Mappings:

Let $X = (X, d)$ and $\tilde{X}=(\tilde{X},d)$ be metric spaces.

A mapping $T: X \to \tilde{X}$ is said to be isometric or isometry if T preserve distance.

Preseve distance mean after applying the mapping the distance is preserve, i.e. for all $x, y \in X$

$$\tilde{d}(T_x, T_y) = d(x, y)$$

Isometric Spaces:

The space X is said to be isometric with space $\tilde{X}$ if there exist a bijective isometry of X onto $\tilde{X}$.

X and $\tilde{X}$ are then called isometric spaces.

Vector Space

Definition:

A vector space (or linear space) over a field K is a nonempty set X of elements x,y,……..(called vectors) together with two algebraic operations.

These operations are called vector addition and multiplication of vectors by scalars, that is, by elements of K.

Vector Addition associates with every ordered pair (x,y) of vectors a vector x+y, called the sum of x and y, in such a way that the following properties hold

Vector addition is commutative and associative.

Vector Space

- ***Subspace***
- ***Basis of a Vector Space***

Subspace:

A subspace of a vector space X is a nonempty subset Y of X such that addition and scalar multiplication are closed in Y.

Two Types of subspaces

- Improper Subspace: If the span of a subspace is equal to that vector space ;
- Proper Subspace: If the span of a subspace is not equal to that vector space

Linear Combination

A linear combination of vectors $x_1, \ldots\ldots\ldots, x_n$ of a vector space X is an axpression of the form

$a_1 x_1 + \ldots\ldots\ldots\ldots a_m x_m$ where the coefficients $a_1, \ldots\ldots\ldots\ldots, a_m$ are any scalars.

Span of a Set:

For any nonempty subset $M \subset X$ the set of all linear combinations of vectors of M is called the span of M.

Written as “span M”.

Obviously, this is a subspace Y of X, and we say that Y is spanned or generated by M.

Linear Independence:

If two vectors have same direction and different in magnitude then on vector is multiple of other which means that one is dependent to other.

If two vectors have not same direction then one vector is independent to other.

Mathematically:

linearly independent.

$$c_1x_1 + c_2x_2 + \ldots\ldots\ldots\ldots\ldots + c_mx_m = 0$$

if and only if all constant are zero

$$c_1 = c_2 = \ldots\ldots\ldots\ldots\ldots = c_m = 0$$

We call $x_1, x_2, \ldots\ldots\ldots\ldots\ldots, x_m$ linearly independent.

linearly dependent.

If vectors are dependent then their coefficients are not equal to 0 as

let

$$x_1 = 2x_2$$

$$\Rightarrow \quad x_1 - 2x_2 = 0$$

Here coefficient 1 뭐2 0 , so x_1 is dependent of x_2 .

Basis of a Vector Space:

As span of M is also a subspace, if the subspace (collection of vectors) is improper subspace (means span of M is equal to that vector space) and linearly independent (coefficients are equal to zero) then that particular subspace is a Basis of a Vector Space.

So, for basis the subspace have to improper subspace and linear independent.

Vector Space

Dimension (definition):

The number of elements in subspace of a basis is called dimension of that vector space.

- ***Dimension***
  - *i. Finite dimensional vector space*
  - *ii. Infinite dimensional vector space*

Norm (definition):

A norm on a (real or complex) vector space X is a real-valued function on X whose value at an $x \in X$ is denoted by $\|x\|$.

Norm metric:

A norm on X defines a metric d on X which is given by

$$d(x, y) = \|x - y\| \qquad \text{where } x, y \in X$$

and is called the metric induced by the norm as this metric depend on norm so we call it metric induced by norm.

Norm Space:

A normed space X is a vector space with a norm defined on it.

A normed space is denoted by $(X, \|\cdot\|)$ or simply by X.

Banach Space:

A Banach space is a complete normed space, (Complete in the metric defined by the norm).

Unit Sphere

The sphere with center 0 and radius 1, S(0; 1), this we define in $\square^2$, but in any metric space

Those points from x whose norm is 1. $\{x \in X \mid \|x\| = 1\}$,

In a normed space X is called the unit sphere. In norm space the collection of all those points which are equal to 1 is called a Unit Sphere.

Subspace (definition)

A subspace Y of a normed space X is a subspace of X considered as a vector space, with the norm obtained by restricting the norm on X to the subset Y.

This norm on Y is said to be induced by the norm on X.

If Y is closed in X, then Y is called a closed subspace of X.

Subspace l^p :

A subspace Y of a Banach space X is a subspace of X considered as a normed space. Hence we do not require Y to be complete.

Theorem :

A subspace Y of a Banach space X is complete if and only if the set Y is closed in X.

Convergence in Normed Spaces.

The metric function is $d(x, y) = \|x - y\|$

For convergence we define as

i): A sequence (x_n) in a normed space X is convergent if X contains an x such that

$$\lim_{n\to\infty} \|x_n - x\| = 0$$

$x_n \to x$, x limit of (x_n)

Now this definition define for Cauchy sequence

ii): A sequence (x_n) in a normed space X is a Cauchy sequence if for every $\varepsilon > 0$ there is an N such that

$$\|x_m - x_n\| < \varepsilon \quad \text{for all } m, n > N$$

Convergence of Infinite Series

A sequence (x_k) is associate with a sequence of partial sum s_n .

$$s_n = x_1 + x_2 + \ldots\ldots\ldots\ldots x_n \quad \text{where } n=1,\ldots\ldots,.$$

If s_n convergent, $s_n \to s$, then

Basis:

In a normed space X is a Cauchy sequence if for every $\varepsilon > 0$ there is an N such that

Elements of basis of $\square^n$ are $e_1, e_2, \ldots\ldots\ldots\ldots, e_n$, such that

$$e_1 = (1, 0\ldots\ldots\ldots\ldots, 0)$$
$$e_2 = (0, 1\ldots\ldots\ldots\ldots, 0)$$
$$.$$
$$.$$
$$e_1 = (0, 0\ldots\ldots\ldots\ldots, 1)$$

Sometimes it is called Canononical basis of $\square^n$.

Elements are spanning and are linearly independent.

Theorem Completion:

Let $X = (x, \|.\|)$ be a normed space then there is a Banach space $\hat{X}$ and an isometry A from X onto a subspace W of $\hat{X}$ which is dense in $\hat{X}$.

The space $\hat{X}$ is unique, except for isometries.

➢ *Theorem (Completeness)*

Theorem

Every finite dimensional subspace Y of a normed space X is complete. In particular, every finite dimensional normed space is complete.

Proof:

➢ ***Theorem (Closedness)***

As we have already proved that every finite dimensional subspace is complete and we also know that a subspace is complete if and only if it is closed.

Theorem

Every finite dimensional subspace Y of a normed space X is closed in X.

This result is true for finite dimensional subspace but for infinite space it is not true.

➢ *Theorem (Equivalent Norms)*

Definition

A norm $\|\cdot\|$ on a vector space X is said to be equivalent to a norm $\|\cdot\|_o$ on X if there are positive numbers a and b such that for all $x \in X$ we have

$$a\|x\|_o \le \|x\| \le \|x\|_o b$$

Compactness and finite Dimension

- ***Lemma (Compactness)***

Definition

A metric space *X* is said to be compact if every sequence in *X* has a convergent subsequence. A subset M of *X* is said to be compact if M is compact considered as a subspace of *X*, that is if every sequence in M has a convergent subsequence whose limit is an element of M.

Lemma (Compactness)

A compact subset M of a metric space is closed and bounded.

Theorem (Compactness)

In a finite dimensional normed space X, any subset $M \subset X$ is compact if and only if M is closed and bounded.

Proof:

We have to prove that compact implies closed and bounded. This we have proved already. Now we prove the converse only. We have to prove only compact (for finite dimensional only).

- ***Theorem (Continuous Mapping)***
- ***Corollary (Maximum and minimum)***

Theorem

Let X and Y be metric spaces and $T: X \to Y$ be a continuous mapping.

Then the image of a compact subset M of X under T is compact.

Proof:

Operator

In the case of vector spaces and, in particular, normed spaced, a mapping is called an operator.

Linear Operator

A linear operator T is an operator such that

i): the domain 👎(T) of T is a vector space and the range *R*(T) lies in a vector space over the same field.

ii): for all $x, y \in D(T)$ and scalar α

Differentiation:

Let *X* be a vector space of all polynomials on [a, b]. A set of polynomial in denoted by *x(t)*

$$Tx(t) = x'(t) \qquad \forall \quad x(t) \in X$$

When we apply T on polynomial x(t) then $x'(t)$ is also a polynomial. So this operator T maps X onto itself. There is no polynomial whose derivative we can't find.

Integration:

Linear operator T for C[a, b] into itself can be defined by

$$Tx(t) = \int_a^t x(\tau)\, d\tau$$

taa τ is just a variable and C[a, b] is collection of all continuous function on a and b.

Matrices:

We denote matrix by capital letter say A. whose elements are in rows and column.

$$A = (\alpha_{jk})$$

Let with r rows and n column we define a linear operator which is

$$T: \square^n \to \square^r$$

Where $\square^n = \{(x_1, \ldots\ldots, x_n) \mid x_i \in \square\}$, in column form so that we use matrices multiplication

$$\begin{bmatrix} x_1 \\ . \\ . \\ x_n \end{bmatrix}$$

Linear Operators

- ***Theorem (Range and Null space)***

Null space is the collection of those elements from the domain on which we apply the operator and the answer is zero.

➢ ***Inverse Operators***

Operator is a mapping whose domain and range is vector space. Particular in norm space. There is also inverse mapping. For inverse operator the same condition is one-to-one and

onto. One-to-one means image of different elements is different. And onto means the range covers all the set of domain. If these two conditions hold then we can define inverse oprator.